



US011491155B2

(12) **United States Patent**
Spargo et al.

(10) **Patent No.:** **US 11,491,155 B2**

(45) **Date of Patent:** **Nov. 8, 2022**

(54) **SALT OF A
PYRIMIDO[6,1-A]ISOQUINOLIN-4-ONE
COMPOUND**

(71) Applicant: **VERONA PHARMA PLC**, Cardiff
(GB)

(72) Inventors: **Peter Lionel Spargo**, Canterbury (GB);
Edward James French, Canterbury
(GB); **Julian Scott Northen**, South
Shields (GB); **John Mykytiuk**,
Houghton-le-Spring (GB)

(73) Assignee: **VERONA PHARMA PLC**, Cardiff
(GB)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **16/580,515**

(22) Filed: **Sep. 24, 2019**

(65) **Prior Publication Data**

US 2020/0016158 A1 Jan. 16, 2020

Related U.S. Application Data

(63) Continuation of application No. 15/549,500, filed as
application No. PCT/GB2016/050313 on Feb. 10,
2016, now Pat. No. 10,463,665.

(30) **Foreign Application Priority Data**

Feb. 11, 2015 (GB) 1502260

(51) **Int. Cl.**
A61K 31/513 (2006.01)
C07D 471/04 (2006.01)

(Continued)

(52) **U.S. Cl.**
CPC **A61K 31/513** (2013.01); **A61K 9/0075**
(2013.01); **A61K 31/185** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC .. C07D 239/00; C07D 487/00; A61K 31/513;
A61K 31/189; A61K 31/519

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

4,581,172 A 4/1986 Kaiser et al.
5,141,936 A 8/1992 Rupp et al.

(Continued)

FOREIGN PATENT DOCUMENTS

WO 00/58308 10/2000
WO 03/105902 A1 12/2003

(Continued)

OTHER PUBLICATIONS

Franciosi et al., "Efficacy and safety of RPL554, a dual PDE3 and
PDE4 inhibitor, in healthy volunteers and in patients with asthma or
chronic obstructive pulmonary disease: findings from four clinical
trials", The Lancet. Respiratory Medicine, Nov. 2013, vol. 1, No. 9,
pp. 714-727.

(Continued)

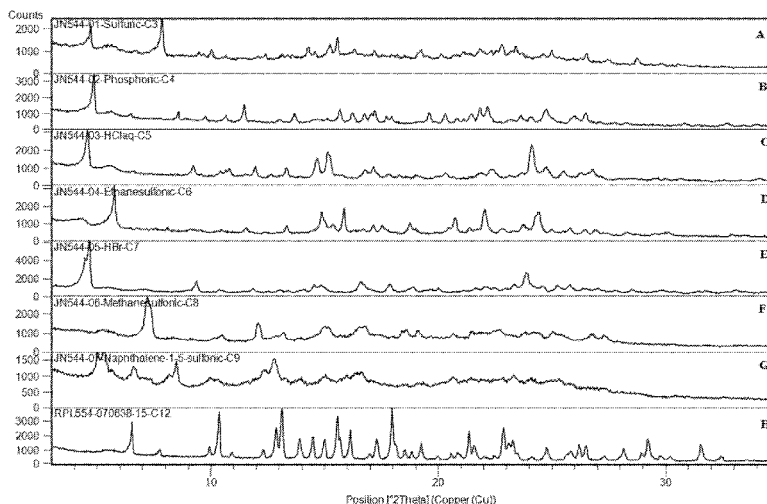
Primary Examiner — San Ming R Hui

(74) *Attorney, Agent, or Firm* — MH2 Technology Law
Group, LLP

(57) **ABSTRACT**

The present invention relates to a pharmaceutically accept-
able acid addition salt of: (i) 9,10-dimethoxy-2-(2,4,6-tri-
methylphenylimino)-3-(N-carbamoyl-2-aminoethyl)-3,4,6,
7-tetrahydro-2H-pyrimido[6,1-a]isoquinolin-4-one
(RPL554); and (ii) ethane-1,2-disulfonic acid, ethanesulfo-
nic acid, methanesulfonic acid, benzenesulfonic acid, p-tolu-
enesulfonic acid, hydrochloric acid, hydrobromic acid,
phosphoric acid or sulfuric acid.

16 Claims, 8 Drawing Sheets



- (51) **Int. Cl.**
A61K 9/00 (2006.01)
A61K 31/185 (2006.01)
A61K 31/519 (2006.01)
A61K 45/06 (2006.01)
- (52) **U.S. Cl.**
 CPC *A61K 31/519* (2013.01); *A61K 45/06* (2013.01); *C07D 471/04* (2013.01); *A61K 2300/00* (2013.01)

2018/0021337 A1 1/2018 Spargo et al.
 2018/0369139 A1 12/2018 Spargo et al.
 2019/0330206 A1 10/2019 Spargo
 2020/0016158 A1 1/2020 Spargo et al.
 2021/0106585 A1 4/2021 Abbott-Banner et al.
 2021/0379053 A1 12/2021 Spargo et al.

FOREIGN PATENT DOCUMENTS

WO 2011/127586 A1 10/2011
 WO 2012/020016 A1 2/2012
 WO 2013/149959 A1 10/2013
 WO 2014/140647 A1 9/2014
 WO 2014/140648 A1 9/2014
 WO 2015/173551 A1 11/2015
 WO 2016/042313 A1 3/2016

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,794,391 B2 9/2004 Oxford et al.
 7,105,663 B2 9/2006 Oxford et al.
 7,378,424 B2 5/2008 Oxford et al.
 8,242,127 B2 8/2012 Oxford et al.
 9,062,047 B2 6/2015 Walker et al.
 9,700,558 B2 7/2017 Walker et al.
 9,717,732 B2 8/2017 Walker et al.
 9,956,171 B2 5/2018 Spargo et al.
 10,463,665 B2 11/2019 Spargo et al.
 10,471,063 B2 11/2019 Walker et al.
 10,710,998 B2 7/2020 Spargo
 10,864,213 B2 12/2020 Abbott-Banner et al.
 10,945,950 B2 3/2021 Spargo et al.
 2003/0036542 A1 2/2003 Oxford et al.
 2004/0171828 A1 9/2004 Oxford et al.
 2004/0176353 A1 9/2004 Oxford et al.
 2008/0206163 A1 8/2008 Oxford et al.
 2012/0302533 A1 11/2012 Oxford et al.
 2013/0225616 A1 8/2013 Walker et al.
 2016/0000790 A1 1/2016 Walker et al.
 2016/0008363 A1 1/2016 Walker et al.
 2017/0112839 A1 4/2017 Abbott-Banner et al.
 2017/0239178 A1 8/2017 Spargo et al.
 2017/0266190 A1 9/2017 Walker et al.

OTHER PUBLICATIONS

Search Report from UK Intellectual Property Office for Application No. GB1502260.1 dated Nov. 3, 2015, 6 pages.
 Stahl et al., Handbook of Pharmaceutical Salts: Properties, Selection, and Use, 2002, VHCA, Zurich, Switzerland and Wiley-VCH, Weinheim, Germany.
 Philip L. Gould, "Salt selection for basic drugs", International Journal of Pharmaceutics, 1986, vol. 33, pp. 201-217.
 Badawy et al., "Microenvironmental pH Modulation in Solid Dosage Forms", Journal of Pharmaceutical Sciences, May 2007, vol. 96, No. 5, pp. 948-959.
 Office Action dated Nov. 19, 2019 for Japanese Application No. 2017-541901, 8 pages with English translation.
 Makoto Ono, "Analysis of Salt Selection of Current Active Pharmaceutical Ingredients (API)", Journal of Pharmaceutical Science and Technology, Japan, 2013, vol. 73, No. 3, pp. 176-182 (See p. 3 (paragraph 2) of the translation of Japanese Office Action dated Nov. 19, 2019 for a concise explanation of the relevance of this document, which is referred to as Reference 6 in the Office Action).

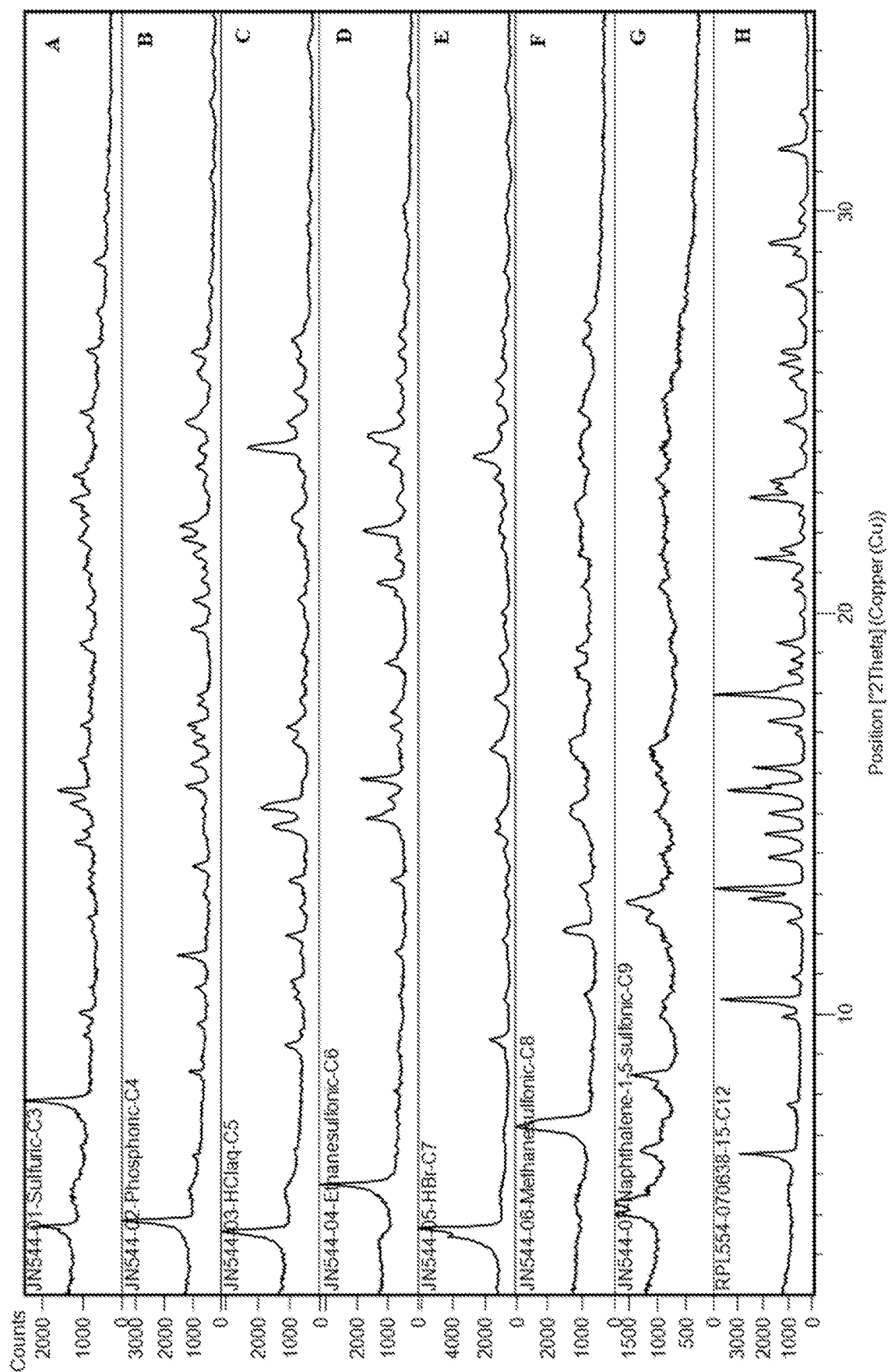


Figure 1

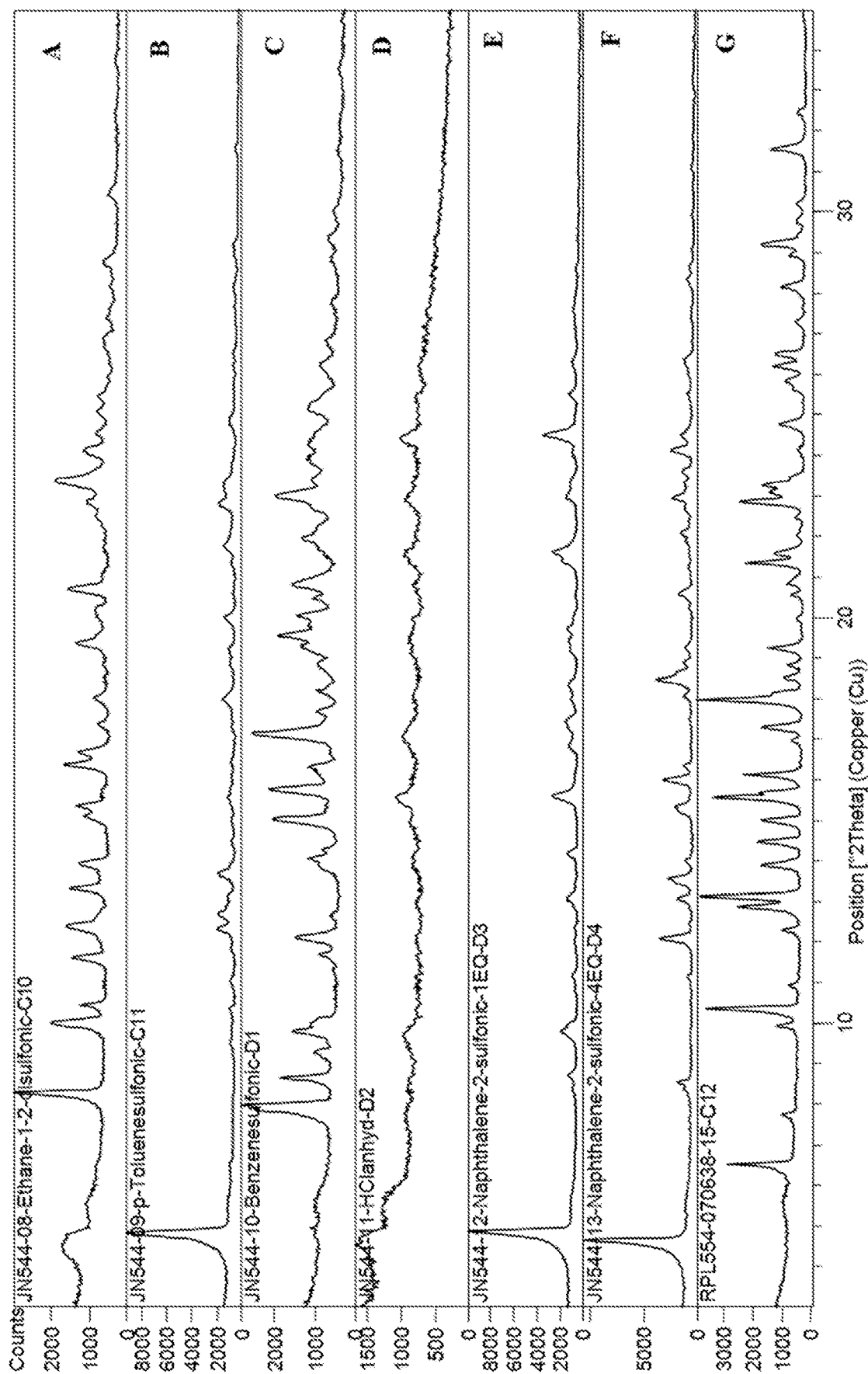


Figure 2

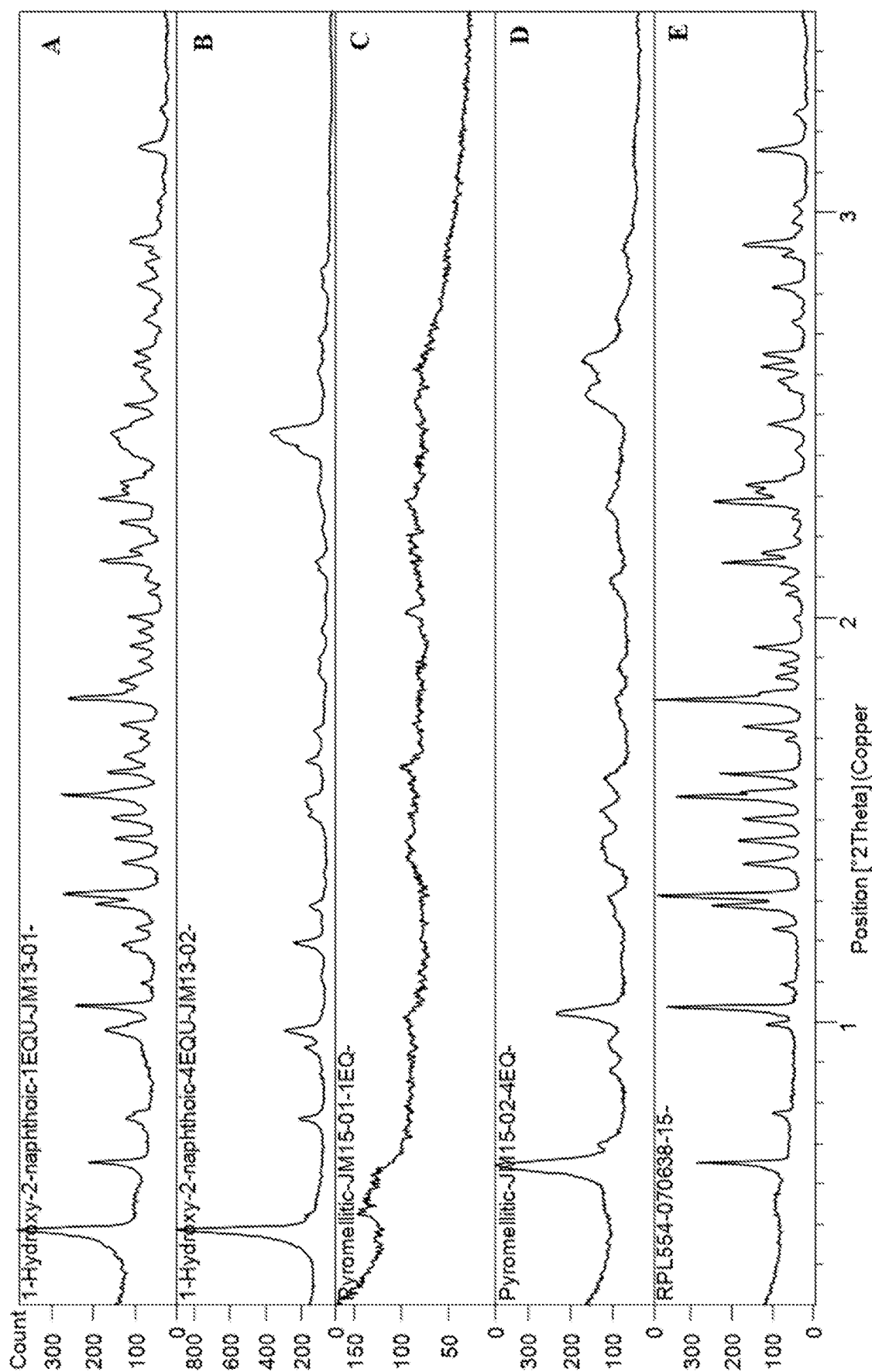


Figure 3

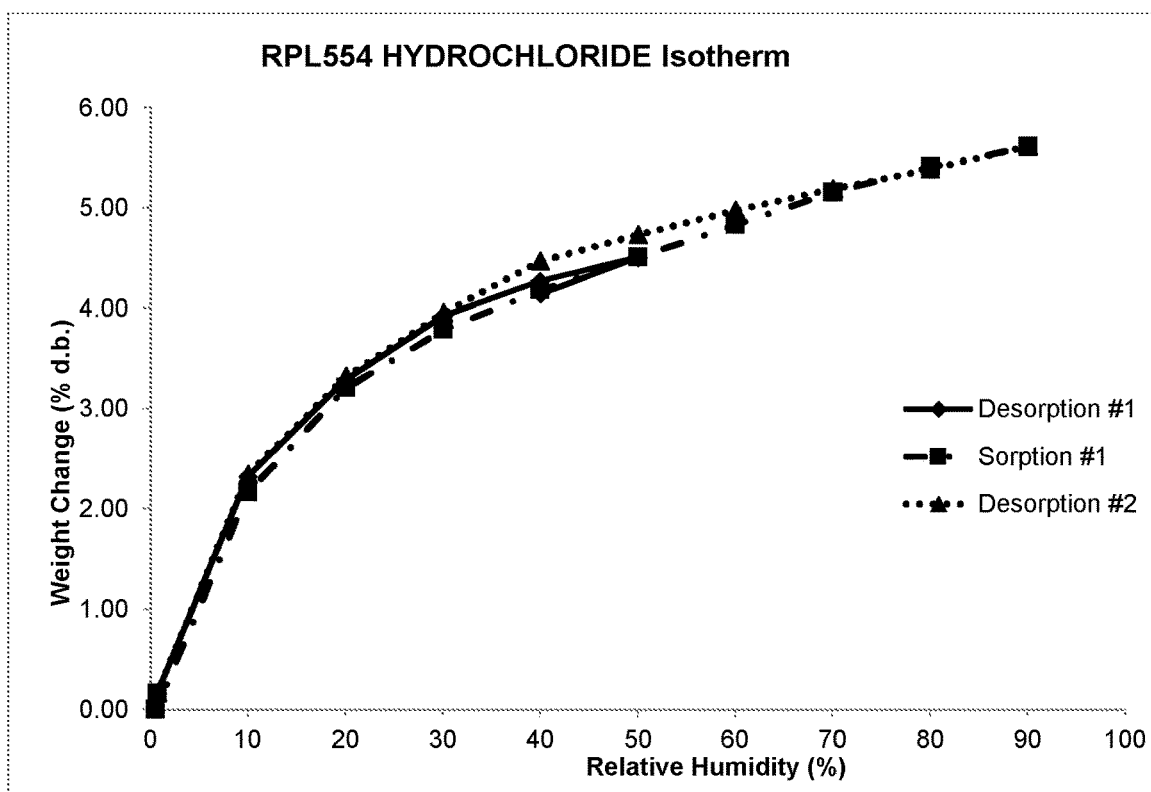


Figure 4

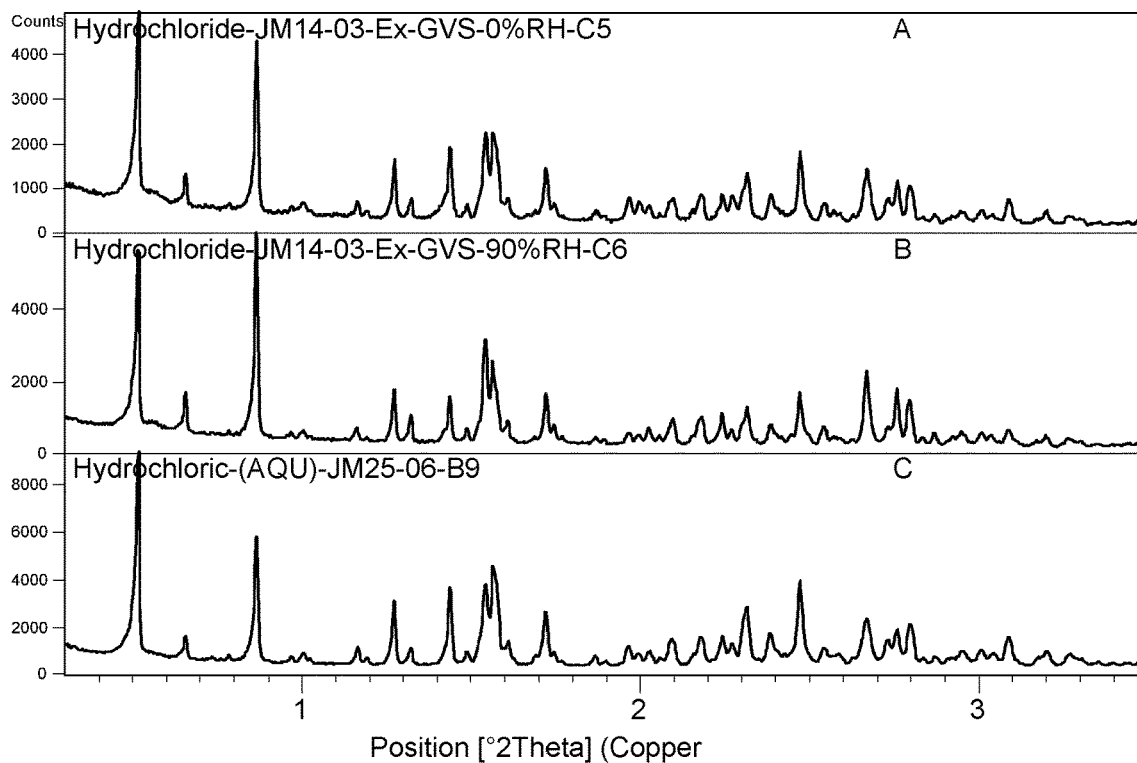


Figure 5

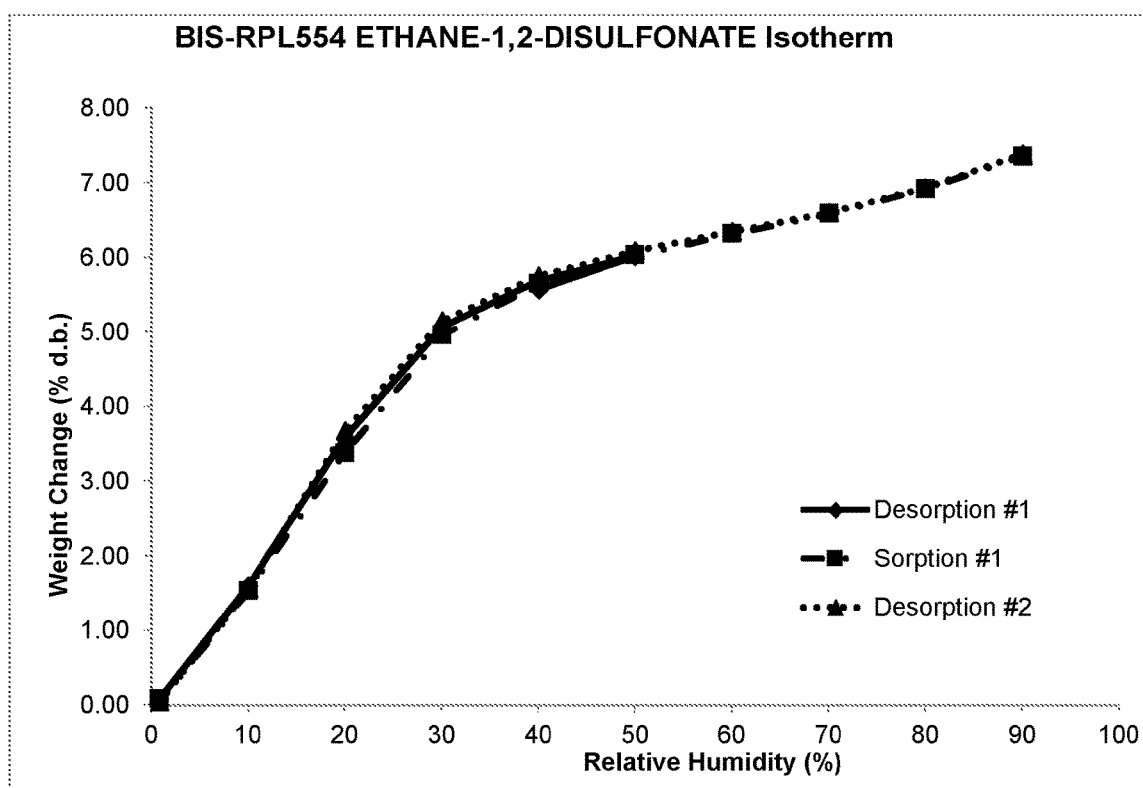


Figure 6

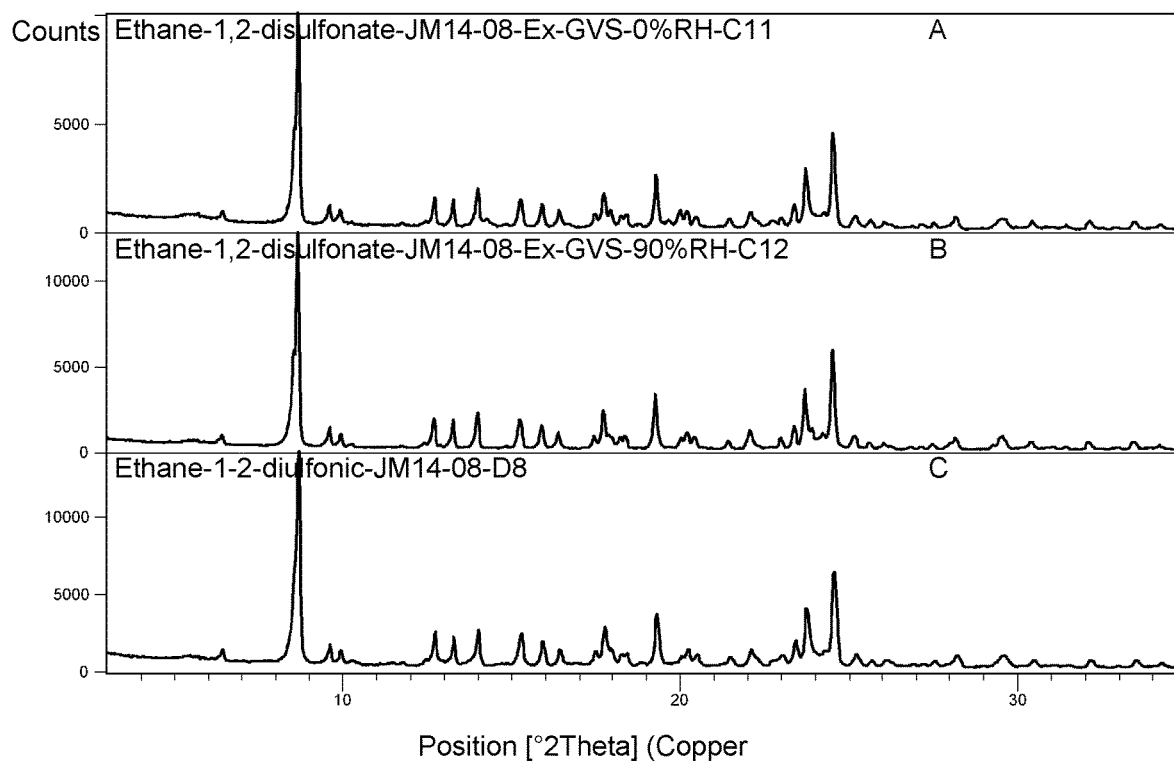


Figure 7

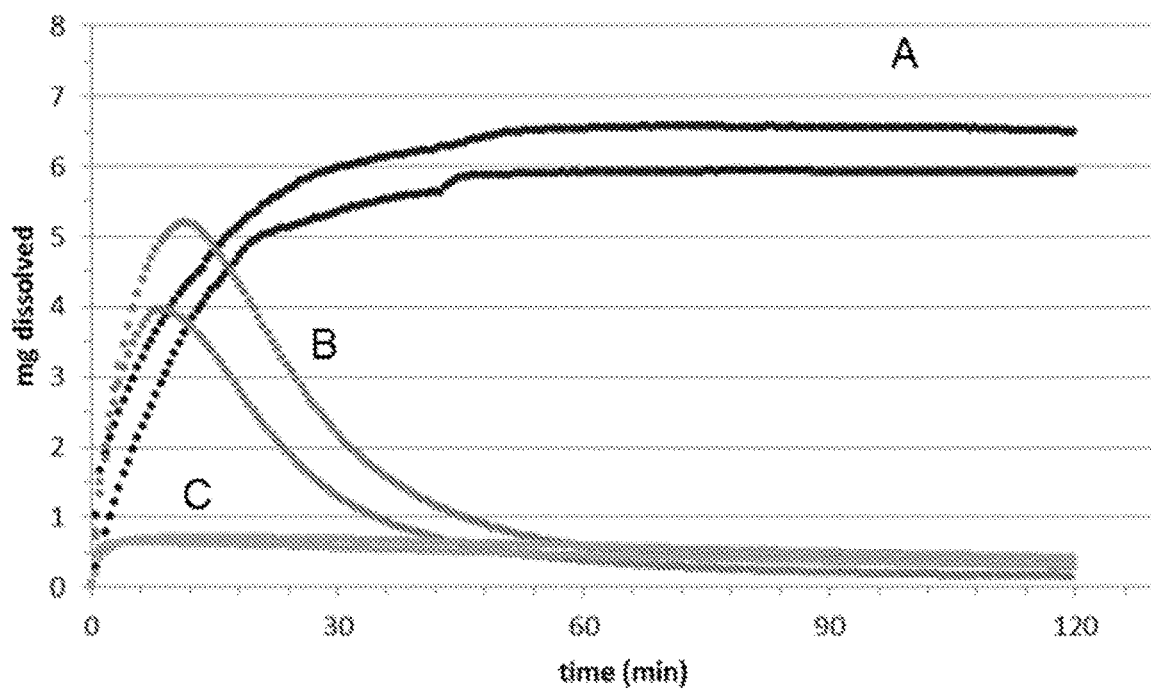


Figure 8

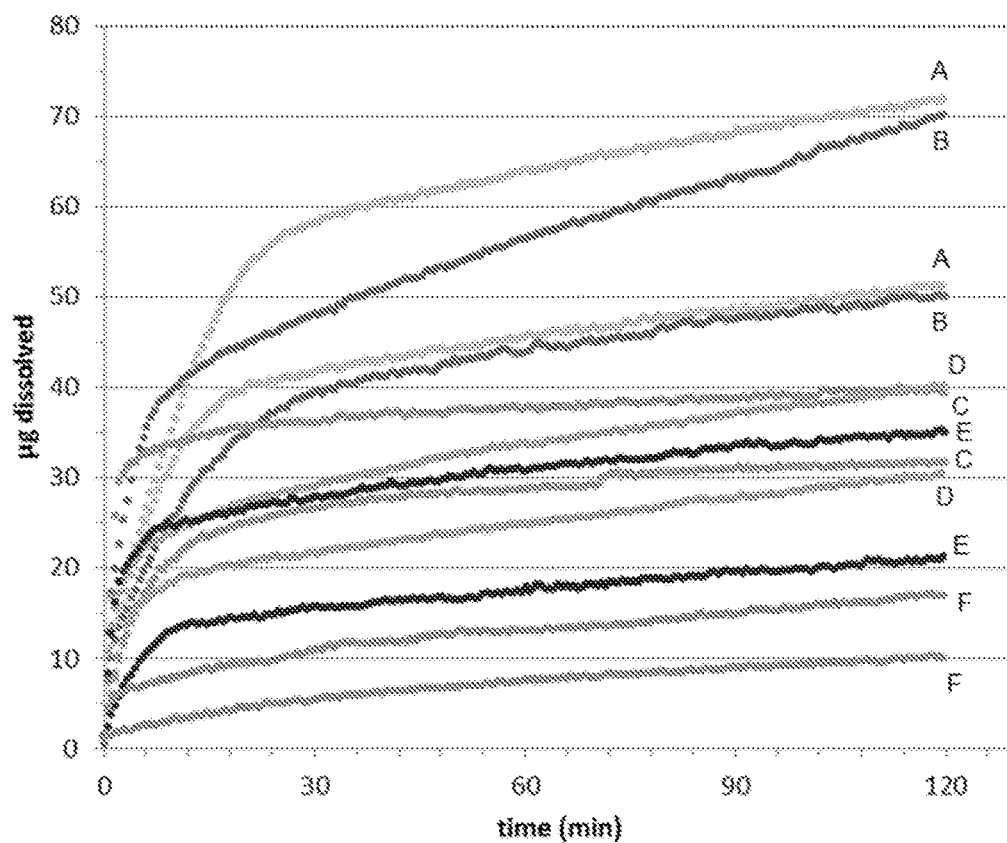


Figure 9

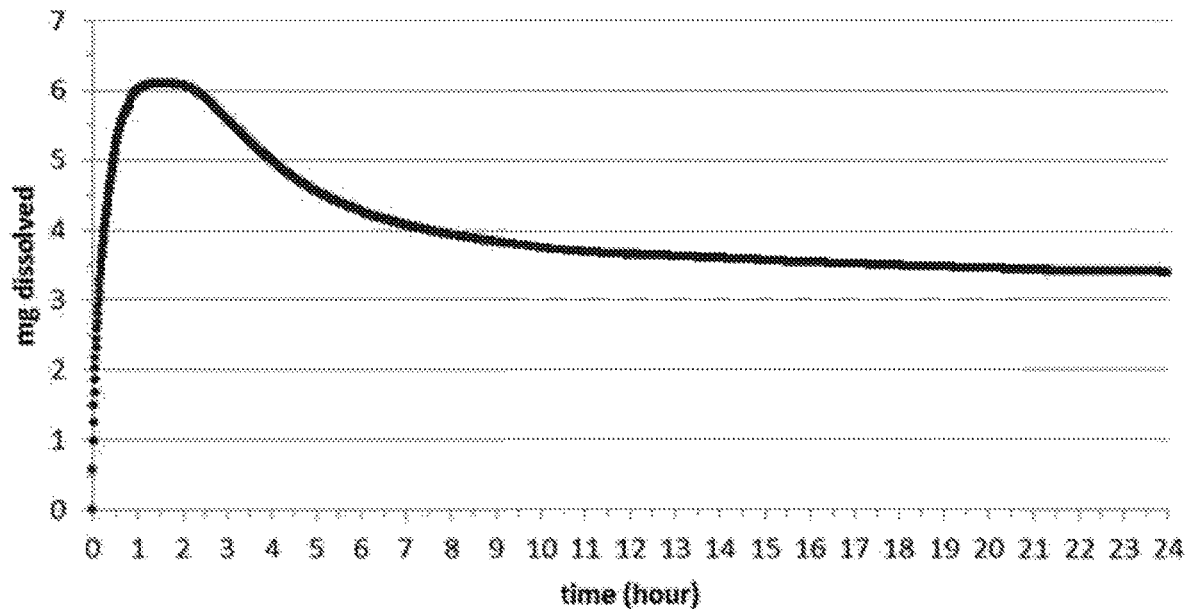


Figure 10

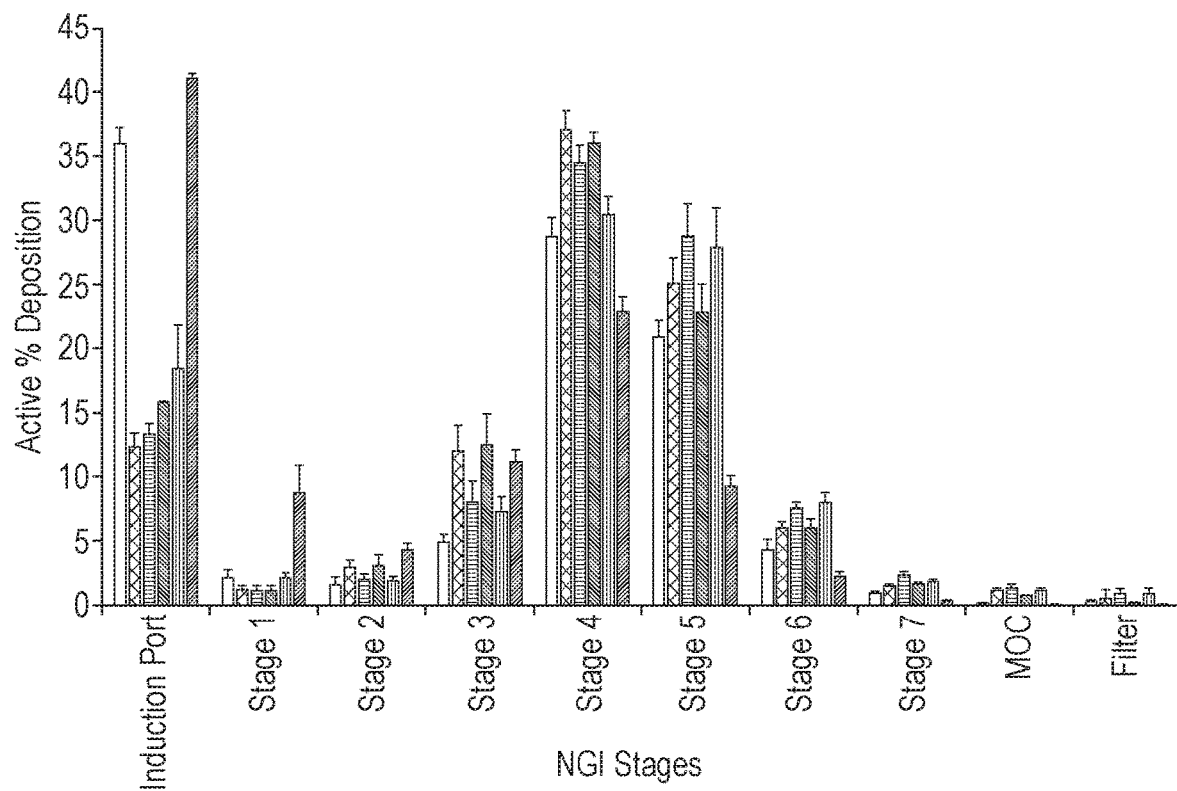


Figure 11

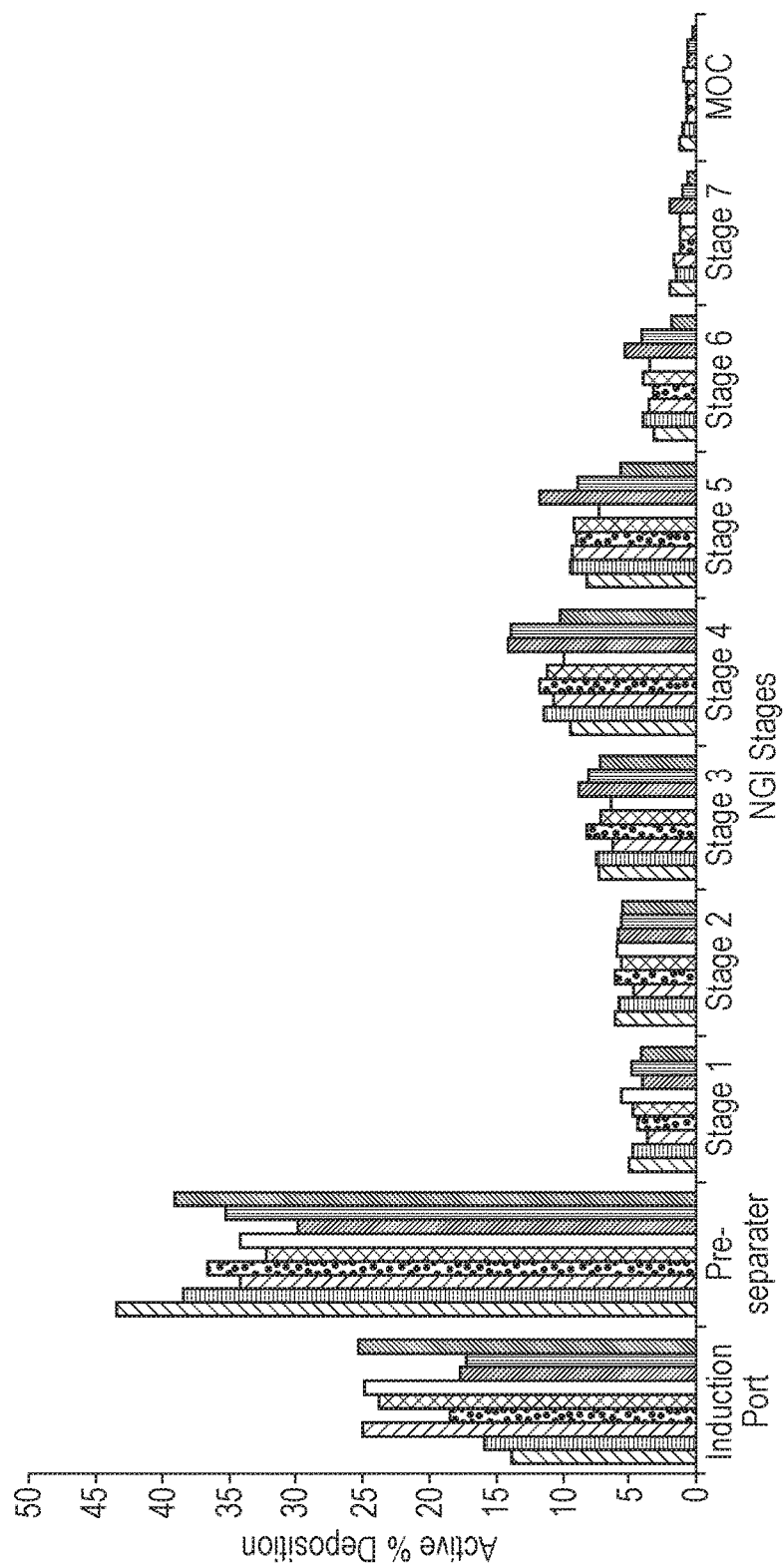


Figure 12

1

**SALT OF A
PYRIMIDO[6,1-A]ISOQUINOLIN-4-ONE
COMPOUND**

CROSS-REFERENCE TO RELATED
APPLICATIONS

This application continuation of U.S. patent application Ser. No. 15/549,500 filed 8 Aug. 2017 (Allowed), which is a U.S. National Stage application of PCT/GB2016/050313, filed 10 Feb. 2016, which claims priority to Great Britain Application No. 1502260.1, filed 11 Feb. 2015, the disclosures of which are hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

The present invention relates to salts of a pyrimido[6,1-A]isoquinolin-4-one compound and pharmaceutical compositions comprising these salts. Medical uses of such salts are also described.

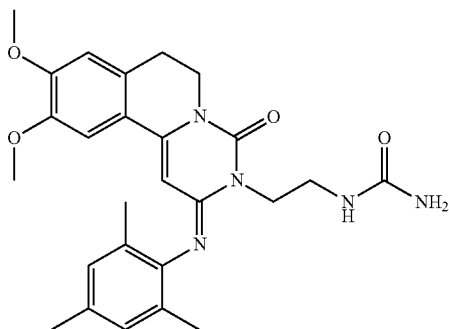
BACKGROUND OF THE INVENTION

A number of factors must be considered when developing an appropriate form of an active pharmaceutical ingredient (API) for formulation for administration to human subjects. These include efficacy, toxicity, stability, solubility/dissolution and acidity/basicity. Different formulations comprising a number of different forms of the API are usually evaluated. The outcomes of such evaluations are unpredictable.

Solubility/dissolution, stability and acidity/basicity can be very important factors to control for an API. Solubility/dissolution must be at an appropriate level to enable delivery of a sufficient dose of the active ingredient to have the desired therapeutic effect in a patient. The solubilities of different salt forms of a therapeutic agent are unpredictable.

The stability of an API must be sufficient to ensure that the product is shelf stable for long periods. Acidity/basicity is of importance for orally administered drugs and inhaled drugs as overly acidic formulations may cause discomfort to subjects upon administration.

RPL554 (9,10-dimethoxy-2-(2,4,6-trimethylphenylimino)-3-(N-carbamoyl-2-aminoethyl)-3,4,6,7-tetrahydro-2H-pyrimido[6,1-a]isoquinolin-4-one) is a dual PDE3/PDE4 inhibitor and is described in WO 00/58308. As a combined PDE3/PDE4 inhibitor, RPL554 has both anti-inflammatory and bronchodilatory activity and is useful in the treatment of respiratory disorders such as asthma and chronic obstructive pulmonary disease (COPD). The structure of RPL554 is shown below.



2

Owing to its applicability in the treatment of respiratory disorders, it is often preferable to administer RPL554 by inhalation. Franciosi et al. disclose a solution of RPL554 in a citrate-phosphate buffer at pH 3.2 (The Lancet: Respiratory Medicine November 2013; 1(9):714-27. DOI: 10.1016/S2213-2600(13)70187-5). The preparation of salts of RPL554 has not been described.

SUMMARY OF THE INVENTION

It is a finding of the present invention that certain salts of RPL554 have highly desirable properties. In particular, specific pharmaceutically acceptable salts of RPL554 have been found to have improved water solubility and improved solubility in appropriately pH buffered solutions. The increased solubility in buffered solutions allows for higher drug loading in solutions of more mild pH, which could lead to greater tolerance of patients to inhalation or oral administration. Certain specific salts of RPL554 have also been found to be crystalline. Crystalline salts are typically more stable than amorphous forms, and are particularly desirable if the RPL554 is to be successfully delivered as a dry powder or in other formulations in which the active ingredient is present in the solid state.

Furthermore, certain salts of RPL554 have been found to have desirable intrinsic dissolution rates which can lead to improved bioavailability. Particular salts of RPL554 have also been found to be well suited to pressurised metered dose and dry powder formulations.

The thermal stabilities of certain salts of RPL554 have also been found to be better than the thermal stabilities of other salts. Such salts are desirable as they are resistant to variation in temperatures without changes in form. Stability to changes in humidity is also desirable.

Accordingly, the present invention provides a pharmaceutically acceptable acid addition salt of:

- (i) 9,10-dimethoxy-2-(2,4,6-trimethylphenylimino)-3-(N-carbamoyl-2-aminoethyl)-3,4,6,7-tetrahydro-2H-pyrimido[6,1-a]isoquinolin-4-one (RPL554); and
- (ii) ethane-1,2-disulfonic acid, ethanesulfonic acid, methanesulfonic acid, benzenesulfonic acid, p-toluenesulfonic acid, hydrochloric acid, hydrobromic acid, phosphoric acid or sulfuric acid.

The invention further provides a pharmaceutical composition comprising a pharmaceutically acceptable acid addition salt of the invention and a pharmaceutically acceptable excipient, carrier or diluent.

The invention further provides a pharmaceutically acceptable acid addition salt of the invention for use in the treatment of the human or animal body.

The invention further provides a pharmaceutically acceptable acid addition salt of the invention for use in the treatment or prevention of a disease or condition selected from asthma, allergic asthma, hay fever, allergic rhinitis, bronchitis, emphysema, bronchiectasis, chronic obstructive pulmonary disease (COPD), adult respiratory distress syndrome (ARDS), steroid resistant asthma, severe asthma, paediatric asthma, cystic fibrosis, lung fibrosis, pulmonary fibrosis, interstitial lung disease, skin disorders, atopic dermatitis, psoriasis, ocular inflammation, cerebral ischaemia, inflammatory diseases and auto-immune diseases.

The invention further provides a method of treating or preventing a disease or condition selected from asthma, allergic asthma, hay fever, allergic rhinitis, bronchitis, emphysema, bronchiectasis, chronic obstructive pulmonary disease (COPD), adult respiratory distress syndrome (ARDS), steroid resistant asthma, severe asthma, paediatric

asthma, cystic fibrosis, lung fibrosis, pulmonary fibrosis, interstitial lung disease, skin disorders, atopic dermatitis, psoriasis, ocular inflammation, cerebral ischaemia, inflammatory diseases and auto-immune diseases in a subject, which method comprises administering to said subject an effective amount of a pharmaceutically acceptable acid addition salt of the invention.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1 shows XRPD patterns of solids isolated from RPL554 salt formation wherein the acid used is: sulfuric (A), phosphoric (B), hydrochloric—aqueous (C), ethanesulfonic (D), hydrobromic (E), methanesulfonic (F) and naphthalene-1,5-disulfonic (G). The XRPD pattern of the RPL554 free base is also shown (H).

FIG. 2 shows XRPD patterns of solids isolated from RPL554 salt formation wherein the acid used is: ethane-1,2-disulfonic (A), p-toluenesulfonic (B), benzenesulfonic (C), hydrochloric—anhydrous (D), 2-naphthalenesulfonic—1 equivalent (E) and 2-naphthalenesulfonic—4 equivalents (F). The XRPD pattern of the RPL554 free base is also shown (G).

FIG. 3 shows XRPD patterns of solids isolated from RPL554 salt formation wherein the acid used is: 1-hydroxy-2-naphthoic—1 equivalent (A), 1-hydroxy-2-naphthoic—4 equivalents (B), pyromellitic—1 equivalent (C) and pyromellitic—4 equivalents (D). The XRPD pattern of the RPL554 free base is also shown (H).

FIG. 4 shows the gravimetric vapour sorption (GVS) profile of RPL554 hydrochloride.

FIG. 5 shows XRPD patterns of RPL554 hydrochloride for 0% humidity (A), 90% humidity (B) and as initially input (C).

FIG. 6 shows the GVS profile of bis-RPL554 ethane-1,2-disulfonate.

FIG. 7 shows XRPD patterns of bis-RPL554 ethane-1,2-disulfonate for 0% humidity (A), 90% humidity (B) and as initially input (C).

FIG. 8 shows the intrinsic dissolution of the sulfate (A), methanesulfonate (B) and ethanesulfonate (C) salts of RPL554.

FIG. 9 shows the intrinsic dissolution of free base (F), ethane-1,2-disulfonate (E), hydrobromide (D), hydrochloride (C), benzenesulfonate (B) and p-toluenesulfonate (A) salts of RPL554.

FIG. 10 shows the intrinsic dissolution of the sulfate salt of RPL554 over 24 hours.

FIG. 11 shows the aerodynamic assessment of selected RPL554 base and salt pMDI formulations (mean values, $n=3$, $\pm$ SD), where the bars from left to right are: free base 134/227 10% EtOH; ethanesulfonate 227; methanesulfonate 134a; benzenesulfonate 134a; sulfate 134a; and ethanedisulfonate 227 10% EtOH.

FIG. 12 shows the aerodynamic assessment of RPL554 free base and salt variant DPI formulations, where the bars from left to right are: free base; ethanesulfonate; p-toluenesulfonate; methanesulfonate; benzenesulfonate; sulfate; hydrochloride; ethane-1,2-disulfonate; and hydrobromide.

DETAILED DESCRIPTION OF THE INVENTION

The RPL554 salts of the invention are pharmaceutically acceptable acid addition salts of RPL554 and ethane-1,2-disulfonic acid, ethanesulfonic acid, methanesulfonic acid,

benzenesulfonic acid, p-toluenesulfonic acid, hydrochloric acid, hydrobromic acid, phosphoric acid or sulfuric acid, or solvates thereof.

Typically, the acid is ethane-1,2-disulfonic acid, ethanesulfonic acid, methanesulfonic acid, benzenesulfonic acid, hydrochloric acid, or sulfuric acid.

An acid addition salt of RPL554 is a salt formed by reaction of RPL554 with an acid. RPL554 is weakly basic compound and may be protonated to form a cationic species which forms a salt with the conjugate base of the acid used.

The acid addition salt of RPL554 may be selected from RPL554 ethane-1,2-disulfonate, RPL554 ethanesulfonate, RPL554 methanesulfonate, RPL554 benzenesulfonate, RPL554 p-toluenesulfonate, RPL554 hydrochloride, RPL554 hydrobromide, RPL554 phosphate, RPL554 sulfate and solvates thereof. Typically, the salt is selected from RPL554 ethane-1,2-disulfonate, RPL554 phosphate, RPL554 methanesulfonate, RPL554 hydrochloride, RPL554 sulfate, and solvates thereof.

Solvates of salts are solid salts which comprise molecules of solvent. The solvent may be any solvent but is often water or an alcohol such as ethanol. The solvate may therefore be a hydrate or an alcoholate such as an ethanolate. The RPL554 salts may be anhydrides, monohydrates, dihydrates or higher hydrates, or non-stoichiometric hydrates.

The stoichiometry of the salts may be one to one, or it may be two to one, or one to two, or indeed any other ratio of acid to base.

The salt of the invention may for instance be RPL554 ethane-1,2-disulfonate or a solvate thereof. The stoichiometry of the RPL554 ethane-1,2-disulfonate salt is typically about 2:1 API:counter ion. The salt of the invention may for instance be RPL554 phosphate or a solvate thereof. The stoichiometry of the RPL554 phosphate salt is typically about 1:1 API:counter ion. The salt of the invention may for instance be RPL554 methanesulfonate or a solvate thereof. The stoichiometry of the RPL554 methanesulfonate salt is typically about 1:1 API:counter ion. The salt of the invention may for instance be RPL554 hydrochloride or a solvate thereof. The stoichiometry of the RPL554 hydrochloride salt is typically about 1:1 API:counter ion. The salt of the invention may for instance be RPL554 sulfate or a solvate thereof. The stoichiometry of the RPL554 sulfate salt is typically about 1:1 API:counter ion.

The stoichiometry of the salts can be determined by techniques known to those skilled in the art, such as ^1H NMR.

The salts of the invention may be produced by any suitable method for producing salts of active pharmaceutical molecules. Typically, the salts are produced by dissolving RPL554 in a solvent, for instance dichloromethane or dimethylformamide, and subsequently adding a solution of the desired acid, for instance a solution of the acid in ethanol or THF. The salt may then be isolated by filtration, or by removing the solvent from the resulting composition, for instance by drying under vacuum. The salt concentrate may be subsequently triturated with an antisolvent, for example methyl tert-butyl ether and collected by filtration, or it may be purified and isolated by other means.

The RPL554 salt of the invention typically has a purity of greater than or equal to about 90%, greater than or equal to about 95% or greater than or equal to about 97%. The percentage may be calculated as area % based on HPLC separation. Thus, the invention provides a solid composition comprising greater than or equal to about 90%, greater than or equal to about 95%, or greater than or equal to about 97%, of a salt of RPL554 as defined herein.

The pharmaceutical composition according to the invention comprises a pharmaceutically acceptable acid addition salt as defined herein and a pharmaceutically acceptable excipient, carrier or diluent. The pharmaceutically acceptable excipient, carrier or diluent may be any suitable pharmaceutically acceptable excipient, carrier or diluent. These are well known to the skilled person.

The diluent may be any pharmaceutically acceptable diluent. The diluent is typically suitable for administration by inhalation. Examples of suitable diluents include water, ethanol and glycerol. The diluent is preferably water. The diluent is preferably sterile. The diluent may alternatively be selected from solid diluents such as lactose, dextrose, saccharose, cellulose, corn starch and potato starch. The diluent may contain buffer components to control the pH. The buffers may be derived from phosphate, citrate or acetate. The diluent may also contain sodium chloride.

The pharmaceutical composition may comprise for instance: lubricants, e.g. silica, talc, stearic acid, magnesium or calcium stearate, and/or polyethylene glycols; binding agents; e.g. starches, arabic gums, gelatin, methylcellulose, carboxymethylcellulose or polyvinyl pyrrolidone; disaggregating agents, e.g. starch, alginic acid, alginates or sodium starch glycolate; effervescing mixtures; dyestuffs; sweeteners; wetting agents, such as lecithin, polysorbates, laurylsulphates; and, in general, non toxic and pharmacologically inactive substances used in pharmaceutical formulations. Such pharmaceutical preparations may be manufactured in known manner, for example, by means of mixing, granulating, tableting, sugar coating, or film coating processes.

Compositions which are liquid dispersions for oral administration may be syrups, emulsions and suspensions. The syrups may contain as carriers, for example, saccharose or saccharose with glycerine and/or mannitol and/or sorbitol.

Compositions which are suspensions or emulsions may contain as carrier, for example a natural gum, agar, sodium alginate, pectin, methylcellulose, carboxymethylcellulose, or polyvinyl alcohol. The suspension or solutions for intramuscular injections may contain, together with the active compound, a pharmaceutically acceptable carrier, e.g. sterile water, olive oil, ethyl oleate, glycols, e.g. propylene glycol, and if desired, a suitable amount of lidocaine hydrochloride.

Solutions for injection or infusion, or for inhalation, may contain as carrier, for example, sterile water or preferably they may be in the form of sterile, aqueous, isotonic saline solutions.

Preferably, the composition of the invention is formulated for administration by inhalation.

The pharmaceutical composition may be a dry powder. Typically, a pharmaceutical composition which is a dry powder is delivered using a dry powder inhaler.

Typically, the dry powder inhaler is a Clickhaler, Novolizer, Certihaler, Diskus, Multihaler, Gyrohaler (Vectura Group plc), Aerolizer, Handihaler or Tubospin (PH&T S.p.A.), Acu-Breathe unit (Respirics, Inc.), Conix (Cambridge Consultants Limited), Miat Monohaler (Cyclohaler), Eclipse (Sanofi-Aventis), e-flex (Microdrug AG), Flowcaps (Hovione), Prohaler (Valois Pharm), DirectHaler (Trimel BioPharma), Single Dose SDD (Manta technologies), Monodose (Miat SpA), TwinCaps (Hovione), GenX (CCL), SkyeHaler (SkyePharma), EasyHaler (Orion Pharma), or Taifun (Akela Pharma Inc.), with Clickhaler, Novolizer, Diskus and Aerolizer being the preferred dry powder inhalers.

Typically, a pharmaceutical composition which is a dry powder is prepared by milling, spray drying, fluidized spray drying, spray congealing, micronization, controlled crystal-

lization, co-crystallization, ultrasound assisted crystallization, freeze drying or particle precipitation of the salt. Typically, the resulting powder has a particle size with a mass median aerodynamic diameter from about 1 μ m to about 10 μ m, preferably from about 3.5 μ m to about 10 μ m, more preferably from about 4 μ m to about 5.5 μ m, or from about 5 μ m to about 10 μ m, or from about 5.5 μ m to about 10 μ m or from about 6 μ m to about 10 μ m. The resulting powder may instead have a particle size with a mass median aerodynamic diameter from about 2 μ m to about 5 μ m, for instance from 2.5 μ m to about 4.5 μ m.

A dry powder pharmaceutical composition may comprise (a) from 50.0 to 99.8 wt % lactose powder and (b) from 0.2 to 50.0 wt % of a salt of RPL554 according to the invention.

A preferred dry powder pharmaceutical composition comprises (a) from 80.0 to 99.8 wt % lactose powder and (b) from 0.2 to 20.0 wt % of a salt of RPL554 according to the invention. For instance, the powder pharmaceutical composition may comprise (a) from 95.0 to 99.8 wt % lactose powder and (b) from 0.2 to 5.0 wt % of a salt of RPL554 according to the invention. In some cases, the content of the salt may be approximately 1.0 wt %. For example, the powder pharmaceutical composition may comprise (a) from 99.2 to 99.8 wt % lactose powder and (b) from 0.2 to 0.8 wt % of a salt of RPL554 according to the invention.

In some cases, the pharmaceutical composition suitable for inhalation is in the form of a composition for a pressurised metered dose inhaler (pMDI). The pharmaceutical composition for administration by pMDI may comprise (a) a salt of RPL554 as defined herein and (b) one or more propellants. The one or more propellants are typically selected from hydrofluoroalkanes such as HFA 134a and HFA 227a. The formulation may further comprise (c) ethanol. The formulation may further comprise (d) a surfactant, for instance Tween 80. For example, the pharmaceutical composition may comprise (a) from 0.1 to 1.0 wt % of a salt of RPL554 as defined herein, (b) from 80.0 to 99.9 wt % of a hydrofluoroalkane, (c) from 0.0 to 19.0 wt % ethanol and (d) from 0.0 to 0.5 wt % Tween 80. In one embodiment, the pharmaceutical composition comprises (a) from 0.1 to 0.9 wt % of a salt of RPL554 as defined herein and (b) from total of HFA 134a and/or HFA 227a. Alternatively, the pharmaceutical composition may comprise (a) from 0.1 to 0.9 wt % of a salt of RPL554 as defined herein, (b) from 88.0 to 90.0 wt % total of HFA 134a and/or HFA 227a, (d) from 0.05 to 0.15 wt % of Tween 80 and (c) ethanol to balance.

In some cases, the pharmaceutical composition is an aqueous solution. Thus, the pharmaceutical composition is typically obtainable by dissolving a pharmaceutically acceptable salt of the invention in an aqueous solution. Dissolving the salt may comprise adding a solid form of the salt to the aqueous solution and agitating the resulting mixture to dissolve at least part of the salt. The resulting mixture may be heated to improve dissolution. pH may also be adjusted appropriately.

The concentration of the salt of the invention in the composition which is an aqueous solution is typically greater than or equal to about 0.1 mg/ml, greater than or equal to about 1 mg/ml or greater than or equal to about 2.5 mg/ml. For instance, the concentration of the salt of the invention in the aqueous solution is typically from about 1 mg/ml to about 50 mg/ml, for instance from about 2 mg/ml to 25 mg/ml or from about 2.5 mg/ml to 10 mg/ml.

Typically, the pharmaceutical composition which is an aqueous solution further comprises one or more buffers. The buffers are pharmaceutically acceptable buffers. Thus, the aqueous solution may be a buffered aqueous solution. The

10. A pharmaceutical composition according to claim 7, which is produced by dissolving the pharmaceutically acceptable acid addition salt of claim 1 in an aqueous solution.

11. A pharmaceutical composition according to claim 10, 5 wherein the aqueous solution is a buffered aqueous solution.

12. A pharmaceutical composition according to claim 11, wherein the buffered aqueous solution comprises a phosphate buffer, a citrate buffer or a citro-phosphate buffer.

13. A pharmaceutical composition according to claim 10, 10 wherein the pH of the pharmaceutical composition is greater than or equal to about 2.5.

14. A pharmaceutical composition according to claim 10, wherein the pharmaceutically acceptable acid addition salt is present at a concentration of greater than or equal to about 15 1.0 mg/ml.

15. A method of treating or preventing a disease or condition selected from asthma, allergic asthma, hay fever, allergic rhinitis, bronchitis, emphysema, bronchiectasis, chronic obstructive pulmonary disease (COPD), adult respi- 20 ratory distress syndrome (ARDS), steroid resistant asthma, severe asthma, paediatric asthma, cystic fibrosis, lung fibrosis, pulmonary fibrosis, interstitial lung disease, skin disorders, atopic dermatitis, psoriasis, ocular inflammation, cerebral ischaemia, inflammatory diseases and auto-immune 25 diseases in a subject, which method comprises administering to said subject an effective amount of the pharmaceutically acceptable acid addition salt of claim 1.

16. The method according to claim 15 wherein the disease or condition is chronic obstructive pulmonary disease 30 (COPD).

* * * * *